The Islamic University of Gaza
Faculty of Engineering
Department of Computer Engineering

MedLink

Weekly Project Progress Report

MedLink Team

Our team is a cohesive and well-integrated unit, combining diverse skills and perspectives to achieve a common goal. Each member contributes unique strengths, which enables effective collaboration and ensures that all aspects of the project are developed with high quality and consistency.

UI/UX Designer

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Front-End Developers

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Back-End Developers

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Week Number

Week 1+2

Week 3

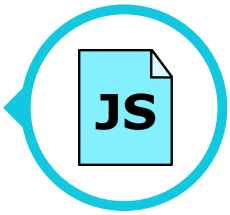
Week 4

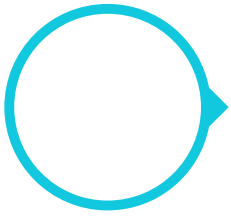
Week 5

Week 6

THIS WEEK

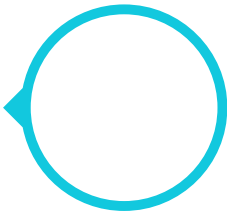
Week7





week 8

Week 9



Work completed this week

To enhance the MedLink platform with dynamic behavior, providing "life" to the static HTML pages and ensuring robust client-side engineering verification through DOM manipulation.

Engineering Verification: Client-side Validation

We prioritized user experience by catching errors before they reach the server, providing immediate visual feedback.

A. Form Integrity & Validation

- Password Matching: In register.js, we implemented logic to compare password fields. If they don't match, the submission is blocked, and the Confirm Password field is highlighted with a red border (#ef4444).
- Required Fields: All critical forms (Login, Register, Request Medicine) prevent empty submissions and provide specific feedback using our custom Toast system.
- Visual Feedback: Fields revert to their default state once the error is corrected, ensuring a "living" form experience.

B. Toast Notification System

We replaced native alert() with a custom MedLinkUI.toast() component in medlink-ui.js:

- Supports success, error, and info types.
- Includes a dynamic progress bar indicating how long the message will stay.
- Uses Glassmorphism and icons for a premium aesthetic.

UI Components & Interactivity

We moved beyond basic browser capabilities to implement custom, high-fidelity UI components.

A. Custom Modals (mlConfirm)

Instead of confirm(), we built a sleek modal system that:

- Blurs the background (backdrop-filter).
- Supports Destructive Actions (e.g., Logout or Delete) by changing icon colors and button styles to red automatically.
- Animates in and out using CSS transitions.

B. Navigation & Sidebars

- Mobile Sidebar/Menu: Implemented a toggle mechanism in citizen-dashboard.js and home.html that handles the visibility of navigation links on small screens.
- Profile Dropdowns: Added logic to open/close profile menus and close them automatically if the user clicks anywhere else on the screen (Outside Click Detection).

C. Interactive Widgets

- Star Rating System: A custom star-rating widget in the pharmacy details page that responds to hover and click events, providing instant visual confirmation of the user's choice.
- Favorite Toggles: "Heart" icons that toggle state (filled vs. outline) instantly, communicating with the localStorage for persistence.

Work completed this week

DOM Manipulation Features

Dynamic Role Switching: The registration form transforms entirely between "Citizen" and "Pharmacy" modes without a page reload.
Search Simulation: A real-time search engine that filters medicine cards as the user types, including "Empty State" handling with a suggestion to "Request Medicine".
Loading States: Every major action (Sign In, Register, Submit Complaint) triggers a fas fa-spinner fa-spin animation on the button to prevent double-clicks and provide feedback.

Features implemented

Structure & Responsiveness (Flexbox/Grid)

Requirement	Status	Key JS File
DOM Manipulation	Completed	register.js, citizen-dashboard.js
Validation	Completed	login.js, register.js
Modals	Completed	medlink-ui.js
Sidebars/Menus	Completed	medlink-ui.js, home.js

Figma Design Link: https://www.figma.com/design/j8pugP9ert6B7c2p8xbjYV/Untitled?node-id=0-1&t=oiR9N0SdGyyGmbEu-1
GitHub Repository Link:https://github.com/alaaeliwa/medlink
GitHub Pages: https://alaaeliwa.github.io/medlink/frontend/

Problems encountered

- Difficulty managing interaction between multiple elements on the same page (DOM complexity).
- Problems with menus closing (Dropdown/Modal) when clicked outside of them.
- Handling of incorrect input cases in forms.
- Double-click button issues.
- Maintaining good performance with increased interactivity.

Solutions applied

- Organize and divide the code into separate JavaScript files for each function.
- Use event listeners thoughtfully (e.g., click outside an element).
- Add clear validation logic before submitting forms.
- Use loading states to prevent duplication.
- Use localStorage to save user states (e.g., Favorites).
- Improve performance by minimizing unnecessary DOM operations.

Plan for next week

- Implement AJAX for dynamic data fetching without page reload.
- Use Fetch/Axios for API communication.
- Add loading indicators during requests.
- Implement instant search functionality.
- Improve overall UX and responsiveness.

Questions for Instructor

- Is the level of JavaScript and DOM manipulation appropriate for a university project?
- Is the code structure (file splitting such as medlink-ui.js, register.js) well-organized?
- Is the level of UI interactivity sufficient, or do we need to add more elements?
- Is using localStorage appropriate at this stage, or would it be better to connect it directly to the backend?

MidLink
Web app